Program Evaluation with Observational Data

Chapter 2 - Selection on observables

Advanced Empirical Research Method

Romuald Méango

MT2024
Department of Economics, University of Oxford

Table of Contents

Introduction and Main Assumptions
Identification Results
Parametric Regression Methods
- Regression methods: linear model
- Regression methods: weighted regression
- Doubly-robust estimator

Nonparametric Regression methods
- Introduction to nonparametric regression
- Nonparametric regression for Treatment effect

Matching
A checklist
Extension 1: Counterfactual Analysis
Extension 2: Double/Debiased Machine Learning (DDML)

Introduction

We are interested in estimating the causal effect of one variable D, on another variable Y.
In Randomized Controlled Trial:

$$(Y_1, Y_0) \perp\!\!\!\perp D$$

In other set up, there is a **selection problem** $\Rightarrow$ biased estimates of the ATE.

$$\mathbb{E}[Y_1 - Y_0] \quad = \quad \mathbb{E}[Y_1|D=1] - \mathbb{E}[Y_0|D=0] + \text{Selection bias} \tag{1}$$

The same holds for the Average Treatment Effect on the Treated (ATT).

$$\mathbb{E}(Y|D=1) - \mathbb{E}(Y|D=0) = \underbrace{\mathbb{E}(Y_1|D=1) - \mathbb{E}(Y_0|D=1)}_{\text{ATT}} + \underbrace{[\mathbb{E}(Y_0|D=1) - \mathbb{E}(Y_0|D=0)]}_{\text{(Selection) bias}}$$

Unconfoundedness

Unconfoundedness (or conditional random assignment) refers to the case where adjusting for differences in a fixed set of covariates removes the selection bias.

Formally, call $\boldsymbol{X}$ as set of covariates:

Assumption (Unconfoundedness / Conditional independence)

$$(Y_0, Y_1) \perp\!\!\!\perp D | \boldsymbol{X} \tag{2}$$

Interpretation: Within a group where people hold the same covariates, assignment to D can be considered as random with respect to the characteristics Y.

Unconfoundedness

Unconfoundedness implies that there is no selection bias, within a sub-group $\boldsymbol{X} = x$.

$$\mathbb{E}(Y_{0i}|D_i = 1, \boldsymbol{X}) - \mathbb{E}(Y_{0i}|D_i = 0, \boldsymbol{X}) = 0$$

$$\mathbb{E}(Y_{1i}|D_i = 1, \boldsymbol{X}) - \mathbb{E}(Y_{1i}|D_i = 0, \boldsymbol{X}) = 0$$

In fact, for identifying ATE or ATT, we only need the weaker assumption:

Assumption (Conditional Mean independence)

The treatment is mean independent of (Y_0, Y_1) given observed covariates $\boldsymbol{X}$, that is:

$$\mathbb{E}(Y_0|\boldsymbol{X}, D) = \mathbb{E}(Y_0|\boldsymbol{X}), \text{ and} \tag{3}$$

$$\mathbb{E}(Y_1|\boldsymbol{X}, D) = \mathbb{E}(Y_1|\boldsymbol{X}). \tag{4}$$

Overlap

Problem:
$\mathbb{E}(Y_1|D=1, \boldsymbol{X}=x)$ is not identified if no one with $\boldsymbol{X}=x$ is treated (or *vice versa*).
Example: PROGRESA for high income households.

The probability $p(\boldsymbol{X}) \equiv \mathbb{P}(D=1|\boldsymbol{X})$ of treatment conditional on an observed random vector $\boldsymbol{X}$ is called the propensity score.

Assumption (Overlap / Common support)

For all $\boldsymbol{x}$ in the support of $\boldsymbol{X}$

$$p(\boldsymbol{x}) < 1 \quad \text{, and} \tag{5}$$

$$p(\boldsymbol{x}) > 0. \tag{6}$$

Example

```r
set.seed(347628)
n = 10000
Simulate three datasets:
 Start with a set of common observables and unobservables
X1 = rnorm(n, 50, 10)
X2 = sample(c(1,0), replace = TRUE, size = n, prob = c(.4,.6))
X3 = rnorm(n, 0, 1) + X1/10
uY = rnorm(n, 0, .2)
uD = rnorm(n, 0, 1)
```

Example

```
#One with selection on observable, but observable uncorrelated with the outcome.
# Hence it does not matter for the ATE
D  = as.numeric(pnorm(uD + .01*X1 - 1 , 0, 1) > .5)
Y  = uY + 1 + 1*D - 0.5*X2  # The ATE is 1.
dfch2data1 = data.frame(X1, X2, X3, D, Y)


#One with selection on observable, and correlated observable
D  = as.numeric(pnorm(uD + .01*X1 + .8*X2 - 1 , 0, 1) > .5)
Y  = uY + 1 + 1*D - 0.5*X2
dfch2data2 = data.frame(X1, X2, X3, D, Y)
```

Example

```
#One with selection on observable, and no overlap

D  = as.numeric(pnorm(uD + .01*X1 - 1 , 0, 1) > .5)
D = 0*as.numeric(X1>50) + D *as.numeric(X1<=50)
Y  = uY + 1 + 1*D - 0.5*X2 -0.01*X1

dfch2data3 = data.frame(X1, X2, X3, D, Y)
```

Example: ATE

```
# Data Frame 1: True ATE - Selection at random
mean(dfch2data1[dfch2data1$D==1,]$Y) - mean(dfch2data1[dfch2data1$D==0,]$Y)

## [1] 1.004336

# Data Frame 2: Biased downard because of unbserved confounder
mean(dfch2data2[dfch2data2$D==1,]$Y) - mean(dfch2data2[dfch2data2$D==0,]$Y)

## [1] 0.8435057

# Data frame 3: Biased upward because of lack of overlap.
mean(dfch2data3[dfch2data3$D==1,]$Y) - mean(dfch2data3[dfch2data3$D==0,]$Y)

## [1] 1.096415
```

Table of Contents

Introduction and Main Assumptions
Identification Results
Parametric Regression Methods
Regression methods: linear model
Regression methods: weighted regression
Doubly-robust estimator
Nonparametric Regression methods
Introduction to nonparametric regression
Nonparametric regression for Treatment effect
Matching
A checklist
Extension 1: Counterfactual Analysis
Extension 2: Double/Debiased Machine Learning (DDML)

Identification of ATE

Selection on observable framework gives us two estimands of the ATE:

Theorem

Under Unconfoundedness and Overlap,

$$ATE \equiv \mathbb{E}(Y_1 - Y_0) = \mathbb{E}_{\boldsymbol{X}}\left[\mathbb{E}(Y|\boldsymbol{X}, D = 1)\right] - \mathbb{E}_{\boldsymbol{X}}\left[\mathbb{E}(Y|\boldsymbol{X}, D = 0)\right].$$

or

$$ATE \equiv \mathbb{E}(Y_1 - Y_0) = \mathbb{E}\left[\frac{\{D - p(\boldsymbol{X})\}\, Y}{p(\boldsymbol{X})\{1 - p(\boldsymbol{X})\}}\right].$$

Proof: See Lecture notes (Core ERM).

Relationship with other parameters

Remark that:

1. Equations 3 and 5 are enough to identify the ATT.
2. Equations 4 and 6 are enough to identify the ATNT.
3. Define:

$$
\begin{aligned}
ATE(\boldsymbol{x}) &= \mathbb{E}(Y_1 - Y_0 | X = \boldsymbol{x}) && (7)\\
ATT(\boldsymbol{x}) &= \mathbb{E}(Y_1 - Y_0 | D = 1, X = \boldsymbol{x}) && (8)\\
ATNT(\boldsymbol{x}) &= \mathbb{E}(Y_1 - Y_0 | D = 0, X = \boldsymbol{x}) && (9)
\end{aligned}
$$

then $ATE(\boldsymbol{x}) = ATT(\boldsymbol{x}) = ATNT(\boldsymbol{x})$. It does not mean that $ATE = ATT = ATNT$!

Relationship with other parameters

4. Restricted supports: Let $\mathscr{X}(d) = \{x : \Pr(D = d|\boldsymbol{x}) > 0\}$ the support in each population and $\mathscr{X} = \mathscr{X}(0) \cap \mathscr{X}(1)$.

$$\text{ATE}(\mathscr{X}) = \mathbb{E}_{\boldsymbol{X} \in \mathscr{X}}\left[\mathbb{E}(Y|\boldsymbol{X}, D = 1)\right] - \mathbb{E}_{\boldsymbol{X} \in \mathscr{X}}\left[\mathbb{E}(Y|\boldsymbol{X}, D = 0)\right].$$

5. If $D \perp\!\!\!\perp (Y_0, Y_1)|\boldsymbol{X}$ then $D \perp\!\!\!\perp (Y_0, Y_1)|p(\boldsymbol{X})$. Hence,

$$ATE(\boldsymbol{x}) = ATE\left(p(\boldsymbol{x})\right).$$

Relationship with other parameters

This setup gives rise to two classes of methods:

- The regression method (parametric or nonparametric):
  - Estimate by regression $\mathbb{E}(Y_d|D=d, \boldsymbol{X}=\boldsymbol{x})$ for all $\boldsymbol{x} \in \mathscr{X}(d)$;
  - Estimate by regression $p(\boldsymbol{X})$ and re-weight observations of treated and untreated.
- The matching method: Compare observations in the control and treatment groups having the same or close characteristics $\boldsymbol{x}$ or propensity score.

Table of Contents

Introduction and Main Assumptions
Identification Results
Parametric Regression Methods
- Regression methods: linear model
- Regression methods: weighted regression
- Doubly-robust estimator

Nonparametric Regression methods
- Introduction to nonparametric regression
- Nonparametric regression for Treatment effect

Matching
A checklist
Extension 1: Counterfactual Analysis
Extension 2: Double/Debiased Machine Learning (DDML)

Linear Model

Assume:

$$Y_{id} = \alpha_d + \boldsymbol{X_i}\beta_d + U_{id}, \text{ with } \mathbb{E}\left(U_{id}|D = d, \boldsymbol{X_i}\right) = 0,$$

where α_d and β_d are deterministic scalars or vectors.

The identifying assumptions are:

1. $\mathbb{E}\left(U_{id}|D = 1, \boldsymbol{X_i}\right) = \mathbb{E}\left(U_{id}|D = 0, \boldsymbol{X_i}\right) = 0$, for $d = 0, 1$;
2. $rank\left(\mathbb{E}\left(\tilde{\boldsymbol{X}}_i'\tilde{\boldsymbol{X}}_i|D_i = d\right)\right)$ is full, with $\tilde{\boldsymbol{X}}_i = (1, \boldsymbol{X}_i)$.

Hence:

$$\begin{aligned} ATT(\boldsymbol{x}) &= \alpha_1 - \alpha_0 + \boldsymbol{x}\left(\beta_1 - \beta_0\right), \text{ and} && (10) \\ ATT &= \alpha_1 - \alpha_0 + \mathbb{E}\left(\boldsymbol{X}|D = 1\right)\left(\beta_1 - \beta_0\right) && (11) \\ ATE &= \alpha_1 - \alpha_0 + \mathbb{E}\left(\boldsymbol{X}\right)\left(\beta_1 - \beta_0\right) && (12) \end{aligned}$$

Linear Model

The linear model (a.k.a index regression model)

$$Y_{id} = \alpha_d + \boldsymbol{X_i}\beta_d + U_{id}, \text{ with } \mathbb{E}\left(U_{id}|D = d, \boldsymbol{X_i}\right) = 0,$$

solves:

1. the dimensionality issue, i.e., is estimable as long as $n_d > dim(\tilde{\boldsymbol{X}})$;
2. the support issue by extrapolation: there can be regions of $\boldsymbol{X}$'s for which $p(\boldsymbol{x}) \in \{0, 1\}$.

with the drawback that:

1. Linear approximation and extrapolation can be inaccurate.

Linear Model - Empirical strategy

Pooled regression:

$$\begin{aligned} Y_i &= Y_{i0} + D_i\left(Y_{i1} - Y_{i0}\right) \\ &= \alpha_0 + \boldsymbol{X}_i\beta_0 + D_i\left(\alpha_1 + \boldsymbol{X}_i\beta_1 - \alpha_0 - \boldsymbol{X}_i\beta_0\right) + U_{i0} + D_i(U_{i1} - U_{i0}) \\ &= \alpha_0 + \boldsymbol{X}_i\beta_0 + D_i\left(\alpha_1 - \alpha_0\right) + D_i\boldsymbol{X}_i\left(\beta_1 - \beta_0\right) + \varepsilon_i, \text{ with } \mathbb{E}\left(\varepsilon_i | X_i, D_i\right) = 0 \end{aligned}$$

OLS regression delivers four types of (consistent and asymptotically normal) estimators:

1. of the intercept: $\hat{\alpha}_0$;
2. of the covariates: $\hat{\beta}_0$;
3. of the treatment variable: $\hat{\alpha}_1 - \hat{\alpha}_0$;
4. of the interaction between the treatment variable and covariates: $\hat{\beta}_1 - \hat{\beta}_0$.

Linear Model - Empirical strategy

"Plug-in" estimates:

$$\widehat{ATE} = (\hat{\alpha}_1 - \hat{\alpha}_0) + \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{x}_i)\left(\hat{\beta}_1 - \hat{\beta}_0\right) \tag{13}$$

$$\widehat{ATT} = (\hat{\alpha}_1 - \hat{\alpha}_0) + \frac{1}{n_T}\sum_{i\in T}(\boldsymbol{x}_i)\left(\hat{\beta}_1 - \hat{\beta}_0\right) \tag{14}$$

Variance is obtained as a consequence of the delta method:

$$\widehat{Var}(\widehat{ATE}) = \begin{pmatrix} 1 & \bar{\boldsymbol{X}} \end{pmatrix} \widehat{Var}\left(\begin{pmatrix} \hat{\alpha}_1 - \hat{\alpha}_0 \\ \hat{\beta}_1 - \hat{\beta}_0 \end{pmatrix}\right) \begin{pmatrix} 1 \\ \bar{\boldsymbol{X}} \end{pmatrix}$$

Linear Model - Empirical strategy

Centering covariates, i.e., use $\boldsymbol{X_i} - \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{x}_i)$ instead of $\boldsymbol{X}_i$, simplifies the procedure

The estimated coefficients are now:

$$\begin{aligned}\hat{\alpha}_d^c &= \hat{\alpha}_d + \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{x}_i)\hat{\beta}_d \\ \hat{\beta}_d^c &= \hat{\beta}_d\end{aligned}$$

Hence:

$$\widehat{ATE} = (\hat{\alpha}_1^c - \hat{\alpha}_0^c) \text{ and } \widehat{Var}\left(\widehat{ATE}\right) = \hat{\sigma}_{\hat{\alpha}_1^c - \hat{\alpha}_0^c}.$$

Question

How do I obtain the ATT?

Example: ATE

```
#library(estimatr)
options(warn=-1)

suppressPackageStartupMessages(library(dplyr))
suppressPackageStartupMessages(library(tidyverse))
#Centering the variables
demeaned.df2<- dfch2data2 %>%
  mutate(x1 = X1 - mean(X1),x2 = X2 - mean(X2),x3 = X3 - mean(X3))
```

Example: ATE

```
#Linear regression model
reg_pooled_model <- lm(Y ~ D+ x1 + x2 + x3, demeaned.df2)
summary(reg_pooled_model)$coefficients

##                  Estimate   Std. Error       t value  Pr(>|t|)
## (Intercept)  7.986128e-01 0.0027214342  293.45292098 0.0000000
## D            1.002407e+00 0.0042109697  238.04655839 0.0000000
## x1          -9.865649e-06 0.0002826891   -0.03489929 0.9721607
## x2          -5.031016e-01 0.0042560053 -118.20980511 0.0000000
## x3           2.853079e-03 0.0019751238    1.44450638 0.1486279
```

Linear Model (and other regression model)

Critical:
Postulated model for $\mathbb{E}(Y|\boldsymbol{X}, D = d) := m(\boldsymbol{X})$ must be the **true regression** of Y on D and $\boldsymbol{X}$.

Estimator for $\widehat{\text{ATE}}_{RA}$ will be biased (inconsistent) if the regression model used is incorrectly specified !

Model-based approaches masks failures of the overlap assumption:
extrapolates value for $\mathbb{E}[Y|D = d, \boldsymbol{X} = \boldsymbol{x}]$ even if there are no individuals in the dataset with $(D = d, \boldsymbol{X} = \boldsymbol{x})$, from values that are actually contained in the dataset.

Weighted regression - Empirical strategy

The key results are:

$$\text{ATE} = \mathbb{E}\left[\frac{\{D - p(\boldsymbol{X})\}\, Y}{p(\boldsymbol{X})\{1 - p(\boldsymbol{X})\}}\right] \text{ and } \text{ATT} = \frac{1}{\Pr(D=1)}\mathbb{E}\left[\frac{\{D - p(\boldsymbol{X})\}\, Y}{\{1 - p(\boldsymbol{X})\}}\right]$$

As before, we use a "Plug-in" estimator. For example:

$$\widehat{\text{ATT}}_n = \frac{1}{\left(\frac{n_T}{n}\right)}\frac{1}{n}\left[\frac{\{D - \hat{p}(\boldsymbol{X})\}\, Y}{1 - \hat{p}(\boldsymbol{X})}\right]$$

For this estimate to be consistent, we need:

1. a consistent estimator $\hat{p}(\boldsymbol{X})$;
2. that $p(\boldsymbol{X})$ is far enough from 1. The denominator cannot be too close to 0. This is related to the support/overlap condition.

Weighted regression - Empirical strategy

Model for propensity score:

$$D_i = I\{\boldsymbol{X}\beta + \varepsilon > 0\},$$

where $Y^* = \boldsymbol{X}\beta + \varepsilon$ is a latent variable, $\boldsymbol{X}$ are explanatory variables, β is the parameter of interest and $F_\varepsilon(.|\boldsymbol{X})$ is the cumulative distribution function (CDF) of errors.

Hence:

$$p(\boldsymbol{x}) = \Pr(\varepsilon < -\boldsymbol{X}\beta) = 1 - F_\varepsilon(-\boldsymbol{X}\beta|\boldsymbol{X})$$

The most popular specifications:

1. Logit: $F_\varepsilon(a|\boldsymbol{X}) = 1/(1+\exp(-a))$;
2. Probit: $F_\varepsilon(a|\boldsymbol{X}) = \Phi(a)$;

Maximum Likelihood Estimator:
$\hat{\beta} = \text{argmax} \sum_{i=1}^{n} D_i \log(1 - F_\varepsilon(-\boldsymbol{X}\beta|\boldsymbol{X})) + (1 - D_i)\log F_\varepsilon(-\boldsymbol{X}_i\beta|\boldsymbol{X})$

Weighted regression

Critical:
Postulated model for $\mathbb{E}(D|\boldsymbol{X})$ must be identical to the **true propensity score**. Estimator $\widehat{\text{ATE}}_{IPW}$ will be biased (inconsistent) if the propensity score model used is incorrectly specified !

Masks failures of overlap: the logit model will never give $\hat{p}(\boldsymbol{X}) = 0$ or 1. However, the estimate will be unstable in the neighbourhood of 0 and 1. It is commonly accepted to trim the data on the propensity score.

Hirano and Imbens (2001) propose to normalise the weights based on the sum of the propensity scores in the treated and control groups and thus makes the weights themselves sum to 1 in each group. It precludes that the weights approach infinity.

Estimation of ATE

It is possible to construct a “doubly-robust” estimator, which is consistent if:

1. the regression model is incorrect but the propensity score model is correct;
2. the regression model is correct but the propensity score model is incorrect;

“Doubly-robust” estimator

“Doubly-robust” estimator

$$\widehat{\text{ATE}}_{DR} \equiv \frac{1}{n}\sum_{i=1}^{n}\left[\frac{D_i Y_i}{\widehat{p}(\boldsymbol{X}_i)} - \frac{[D_i - \widehat{p}(\boldsymbol{X}_i)]}{\widehat{p}(\boldsymbol{X}_i)}\hat{\mu}_1(\boldsymbol{X}_i)\right]$$

$$-\frac{1}{n}\sum_{i=1}^{n}\left[\frac{(1-D_i)Y_i}{1-\widehat{p}(\boldsymbol{X}_i)} - \frac{[D_i - \widehat{p}(\boldsymbol{X}_i)]}{1-\widehat{p}(\boldsymbol{X}_i)}\hat{\mu}_0(\boldsymbol{X}_i)\right]$$

Estimation of ATE

Take the first term, say $\hat{\mu}_{1,DR}$. By the Law of Large Numbers, it estimates:

$$
\begin{aligned}
&\mathbb{E}\left[\frac{DY}{\pi(\boldsymbol{X})} - \frac{D-\pi(\boldsymbol{X})}{\pi(\boldsymbol{X})} m_1(\boldsymbol{X})\right] \\
&= \mathbb{E}\left[\frac{DY_1}{\pi(\boldsymbol{X})} - \frac{D-\pi(\boldsymbol{X})}{\pi(\boldsymbol{X})} m_1(\boldsymbol{X})\right] \\
&= \mathbb{E}\left[\frac{DY_1}{\pi(\boldsymbol{X})} - \frac{(D-\pi(\boldsymbol{X}))Y_1}{\pi(\boldsymbol{X})} + \frac{(D-\pi(\boldsymbol{X}))Y_1}{\pi(\boldsymbol{X})} - \frac{D-\pi(\boldsymbol{X})}{\pi(\boldsymbol{X})} m_1(\boldsymbol{X})\right] \\
&= \mathbb{E}\left[Y_1 + \frac{D-\pi(\boldsymbol{X})}{\pi(\boldsymbol{X})}(Y_1 - m_1(\boldsymbol{X}))\right]
\end{aligned}
$$

where $m_1(\boldsymbol{X})$ and $\pi(\boldsymbol{X})$ are the postulated models.

Estimation of ATE

Take the first term, say $\hat{\mu}_{1,DR}$. By the Law of Large Numbers, it estimates:

$$\mathbb{E}\left[\frac{DY_1}{\pi(\boldsymbol{X})} - \frac{D-\pi(\boldsymbol{X})}{\pi(\boldsymbol{X})} m_1(\boldsymbol{X})\right] = \mathbb{E}[Y_1] + \mathbb{E}\left[\frac{D-\pi(\boldsymbol{X})}{\pi(\boldsymbol{X})}(Y_1 - m_1(\boldsymbol{X}))\right]$$

For $\hat{\mu}_{1,DR}$ to estimate $\mathbb{E}[Y_1]$, the additional term must be 0.

Question

Prove that the additional term equals 0 if either:

1. $\mathbb{E}(Y_1|\boldsymbol{X}) = m_1(\boldsymbol{X})$ (even if $p(\boldsymbol{X}) \neq \pi(\boldsymbol{X})$),
2. $p(\boldsymbol{X}) = \pi(\boldsymbol{X})$ (even if $\mathbb{E}(Y_1|\boldsymbol{X}) \neq m_1(\boldsymbol{X})$).

Estimation of ATE

Practice - weighted regression

Table of Contents

Introduction and Main Assumptions
Identification Results
Parametric Regression Methods
- Regression methods: linear model
- Regression methods: weighted regression
- Doubly-robust estimator

Nonparametric Regression methods
- Introduction to nonparametric regression
- Nonparametric regression for Treatment effect

Matching
A checklist
Extension 1: Counterfactual Analysis
Extension 2: Double/Debiased Machine Learning (DDML)

Why and when is OLS inadequate? (Imbens, 2015)

One can write the OLS estimator of $\mathbb{E}[Y_{0i}|D_i = 1]$ as :

$$\widehat{\mathbb{E}[Y_{0i}|D_i = 1]} = \frac{1}{N_c} \sum_{i:D_i=0} \omega_i Y_i,$$

where, for control unit i the weight ω_i is :

$$\omega_i = 1 - (X_i - \bar{X}_c)\frac{\bar{X}_c - \bar{X}_t}{\overline{X^2} - \bar{X}\bar{X}}$$

The weights average to one over the control sample.

Imbens, 2015

When $\bar{X}_c = \bar{X}_t$, $\omega_i = 1$, $\forall i$, so that $\mathbb{E}[\widehat{Y_{0i}|D_i = 1}] = \bar{Y}_c$; e.g., in a randomized experiment.

Relatively extreme weights are common if $\bar{X}_c$ and $\bar{X}_t$ are very different.

Example: Lalonde (1986) Data. X_i: earnings in 1975.

- $\omega_i = 2.8091 - 0.0949 X_i$.
- largest positive weight is 2.08, for control unit with close to zero income.
- Most negative weight is -12.05, for $X_i = \$156.7$.

Imbens, 2015

"I would argue that whether an individual with earnings equal to \$156,700 makes \$100,000 or \$300,000 in 1978 should make ***no difference*** *for a prediction for the average value of* Y_{0i} *for the trainees in the program, given that the maximum value for* $[X_i]$ *for trainees is 25K"*

"The problem is that the least squares estimates take the linearity in the linear regression model ***very seriously****, and given linearity, observations on units with* ***extreme values for*** $[X_i]$ ***are in fact very informative****."*

"[Parametric] regression methods are fundamentally not robust to the substantial differences between treatment and control groups."

Nonparametric Regression: an Introduction

Nonparametric CDF estimation

How to estimate a cumulative distribution function (CDF)?

$$F(x) = \Pr(X \leq x)$$

Nonparametric CDF estimation

How to estimate a cumulative distribution function (CDF)?

$$F(x) = \Pr(X \leq x)$$

Empirical cdf:

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} I\{X_i \leq x\} \xrightarrow{a.s.} F(x)$$

An estimator of the probability density function $dF(x)/dx$ cannot be constructed since this function as a step function is not differentiable everywhere.

Nonparametric CDF estimation

Idea:

Smoothing: Let $H(.)$ be a smooth function increasing from 0 to 1. Let:

$$F_{nh}(x) = \frac{1}{n}\sum_{i=1}^{n} H\left(\left(x - X_i\right)/h_n\right)$$

When the *bandwith* $h_n \to 0$: $H\left(\left(x - X_i\right)/h_n\right) \to I\left\{X_i \leq x\right\}$.

Nonparametric Density estimation

Let $K(u) = dH(u)/du$ be a Kernel function.

- $K(u)$ is bounded;
- $|u|\,|K(u)| \to 0$ when $|u| \to 0$;
- $\int K(u)du = 1$;
- $\int K(u)^2 du < \infty$;
- $\int |K(u)|\, du < \infty$;

Example: Epaechnikov:

$$K(u) = \frac{3}{4}\left(1 - u^2\right) I\{|u| \leq 1\}$$

library(evmix)

plot(function(x) kdepanechnikov(x), -2, 2)

Other Examples:
`https://search.r-project.org/CRAN/refmans/evmix/html/kernels.html`

Nonparametric Density estimation

Kernel Estimator:

$$f_{nh}(x) = \frac{dF_{nh}(x)}{dx} = \frac{1}{nh_n} \sum_{i=1}^{n} K\left((x - X_i)/h_n\right)$$

If x of dimension d:

$$K_d(u) = \prod_{m=1}^{d} K(u_m), \text{ and}$$

$$f_{nh}(x) = \frac{1}{n(h_n)^d} \sum_{i=1}^{n} K_d\left((x - X_i)/h_n\right)$$

Nonparametric regression

The conditional expectation is $\mathbb{E}(Y|X=x)$ almost surely equal to:

$$\underset{m\in\mathscr{G}}{\operatorname{argmin}}\ \mathbb{E}(Y-m(X))^2$$

The Kernel estimator $m_{nh}(x)$ of $m(x)$ is obtained by replacing the expectation in the minimization problem by its kernel estimator:

$$\underset{m}{\operatorname{argmin}}\ \sum_{i=1}^{n}(y_i-m)^2 K_d\left((x-X_i)/h_n\right)$$

Solving this program yields the estimator:

$$m_{nh}(x)=\frac{\sum_{i=1}^{n} y_i K_d\left((x-X_i)/h_n\right)}{\sum_{i=1}^{n} K_d\left((x-X_i)/h_n\right)}$$

Nonparametric regression - Inference

Under regalurity assumptions on the conditional expectation:

$$\underbrace{\sqrt{n(h_n)^d}}_{\text{Slower as usual}}\left(m_{nh}(x) - m(x) - \underbrace{h_n^2 b(x)}_{\text{Asymptotic Bias}}\right) \xrightarrow[n\to\infty]{d} \mathcal{N}\left(0, \underbrace{v(x)}_{\text{Complicated object}}\right)$$

Basic tradeoff between bias and variance of estimates:

- The smaller h_n is, the larger the variance: the estimation is too noisy.
- The larger h_n is, the larger the bias: the estimation is not 'local' enough.

The 'optimal' h_n is the one which minimizes the sum of the variance and the square of the bias (the mean square error).

Nonparametric regression for Treatment effect

Table: Consistent estimators

Quantity	Estimator
$p(x)$	$p_{nh}(x) = \frac{\sum_{i=1}^{n} d_i K_d\left((x - x_i)/h_n\right)}{\sum_{i=1}^{n} K_d\left((x - x_i)/h_n\right)}$
$\mathbb{E}(Y_i \vert D_i = d, X_i = x)$	$m_{nh}^{(d)}(x) = \frac{\sum_{i=1, d_i=d}^{n} y_i K_d\left((x - X_i)/h_n\right)}{\sum_{i=1, d_i=d}^{n} K_d\left((x - X_i)/h_n\right)}$
	or $\mu_{nh}^{(0)}(x) = \frac{\sum_{i=1}^{n} y_i (1 - d_i) K_d\left((x - X_i)/h_n\right)}{\sum_{i=1}^{n} K_d\left((x - X_i)/h_n\right)} / (1 - p_{nh}(x))$
	and $\mu_{nh}^{(1)}(x) = \frac{\sum_{i=1}^{n} y_i d_i K_d\left((x - X_i)/h_n\right)}{\sum_{i=1}^{n} K_d\left((x - X_i)/h_n\right)} / p_{nh}(x)$
$ATT(X)$	$\widehat{ATT(x)}_{nh} = m_{nh}^{(1)}(x) - m_{nh}^{(0)}(x)$
ATT	$\widehat{ATT}_{nh} = \frac{1}{n_T} \sum_{i=1}^{n} D_i \widehat{ATT(x_i)}_{nh}$
ATE	$\widehat{ATE}_{nh} = \frac{1}{n} \sum_{i=1}^{n} \widehat{ATT(x_i)}_{nh}$

Nonparametric regression - Alternative

Instead of fitting an average, one can fit a straight line,

$$\underset{m,\beta}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - m - x\beta\right)^2 K_d\left(\left(x - X_i\right)/h_n\right)$$

or higher order polynomials.

They deliver directly the estimate for $E(Y|D=d, X=x)$. For example, the local linear regression on the sample of treated, i such that $D=1$, delivers: $\hat{E}(Y|D=1, X=x)$.

R packages for Nonparametric regression

```
##Local Averaging (Friedman's \Super Smoother"):
#supsmu(x, y, wt =, span = "cv", periodic = FALSE, bass = 0, trace = FALSE)

##Local Regression (Cleveland's LOESS):
#loess(formula, data, weights, subset, na.action, model = FALSE,
#           span = 0.75, enp.target, degree = 2,
#            parametric = FALSE, drop.square = FALSE, normalize = TRUE,
#            family = c("gaussian", "symmetric"),
#        method = c("loess", "model.frame"), control = loess.control(...), ...)

##Kernel Regression (Nadaraya and Watson's Kernel Smoother):
#ksmooth(x, y, kernel = c("box", "normal"), bandwidth = 0.5,
#           range.x = range(x), n.points = max(100L, length(x)), x.points)
```

Nonparametric regression

Nonparametric regression generally suffers from the so-called 'curse of dimensionality'. Because it weights down/ discards observations far from the value of interest, we are sometimes left with sparsely populated bins.

You shouldn't use these methods for less than 15 observations in a bin; optimally you should apply them for 50 or more.

To go further: `http://users.stat.umn.edu/~helwig/notes/smooth-notes.html`

Estimation of ATE

Practice - Introduction to nonparametric regression

Table of Contents

Introduction and Main Assumptions
Identification Results
Parametric Regression Methods
- Regression methods: linear model
- Regression methods: weighted regression
- Doubly-robust estimator

Nonparametric Regression methods
- Introduction to nonparametric regression
- Nonparametric regression for Treatment effect

Matching
A checklist
Extension 1: Counterfactual Analysis
Extension 2: Double/Debiased Machine Learning (DDML)

Imbens, 2015

"I would argue that whether an individual with earnings equal to $156,700 makes $100,000 or $300,000 in 1978 should make ***no difference*** *for a prediction for the average value of* Y_{0i} *for the trainees in the program, given that the maximum value for* $[X_i]$ *for trainees is 25K"*

"If, [...] one matched all the treated units, the control units with $[X_i]$ *exceeding 25K would receive zero weight in the estimation, and in fact few control units with* $[X_i]$ *between 18 and 25 would receive positive weights.*

"Matching is therefore by design robust to the presence of such units."

Matching

Idea: compare the outcome of a treated with the outcome of very similar members of the control group.

Similarity is here measured by the distance between covariates.

- Euclidian distance :

$$d_E(x,z) = (x-z)'(x-z)$$

- Mahalanobis distance:

$$d_M(x,z) = (x-z)'(\Omega_X^{-1})(x-z)$$

- Diagonal version of the Mahalanobis distance:

$$d_{AI}(x,z) = (x-z)'diag(\Omega_X^{-1})(x-z)$$

Matching

Formally, given a sample $\{(Y_i, X_i, D_i)\}_{i=1}^N$, let $I_m(i)$ be the index I that satisfies $D_I \neq D_i$ and

$$\sum_{j|D_i \neq D_j} 1\{||X_j - X_i|| \leq ||X_I - X_i||\} = m$$

$I_m(i)$: the m-th closest to unit i in opposite treatment group.

Define: $\mathscr{J}_M(i) = \{I_1(i), \ldots, I_M(i)\}$.

We use this set to impute the unobserved potential outcome variable.

Matching

Define the imputed potential outcomes as:

$$\hat{Y}_{0i} = \begin{cases} Y_i & \text{if } D_i = 0 \\ \frac{1}{M}\sum_{j \in \mathscr{J}_M(i)} Y_j & \text{if } D_i = 1 \end{cases}$$

$$\hat{Y}_{1i} = \begin{cases} \frac{1}{M}\sum_{j \in \mathscr{J}_M(i)} Y_j & \text{if } D_i = 0 \\ Y_i & \text{if } D_i = 1 \end{cases}$$

The simple matching estimator is then:

$$\widehat{ATE}_M^{sm} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{Y}_{1i} - \hat{Y}_{0i}\right)$$

Matching

Issues with Matching:

- Bias of order $O(N^{-1/K})$, where K is the dimension of covariates.
- Is important in large sample if $K \geq 2$ (and dominates variance asymptotically if $K \geq 3$).
- Not efficient, but efficiency loss is small.
- Easy to implement and robust.
- For the variance calculation: bootstrap does not work, subsampling works better.

Imbens, 2015

"I provide some advice for researchers estimating average treatment effects in practice, based on my personal view of this literature."

Estimation in a three-step strategy:

1. Assess Overlap;
   - Compare summary statistics,
   - Drop some observations if necessary to ensure overlap.
2. Assess unconfoundedness;
   - Compute Pseudo-ATE for a pre-determined outcome.
3. Estimation;
   - Matching with bias adjustment,
   - or Blocking with regression.

Imbens, 2015

Assessing Overlap:

Inspection of the normalized difference:
For some element $X_{i,k}$ of the covariate vector x_i:

$$\Delta_{X,k} = \frac{\bar{X}_{t,k} - \bar{X}_{c,k}}{\sqrt{\left(S^2_{X,t,k} + S^2_{X,c,k}\right)/2}}$$

where

$$\bar{X}_{c,k} = \frac{1}{N_c}\sum_{i:D_i=0} X_{i,k}, \quad \bar{X}_{t,k} = \frac{1}{N_t}\sum_{i:D_i=1} X_{i,k}$$
$$S^2_{X,c,k} = \frac{1}{N_c-1}\sum_{i:D_i=0}\left(X_{i,k} - \bar{X}_{c,k}\right)^2; S^2_{X,t,k} = \frac{1}{N_t-1}\sum_{i:D_i=1}\left(X_{i,k} - \bar{X}_{t,k}\right)^2$$

Imbens, 2015

Assessing Overlap:

The normalized difference is more useful than the t-stat, which defined by:

$$\frac{\bar{X}_{t,k} - \bar{X}_{c,k}}{\sqrt{\left(S^2_{X,t,k}/N_t + S^2_{X,c,k}/N_c\right)}}$$

When the sample is large, small differences are statistically significant even if they are **substantively small**.

If the normalized difference is large (≥ 0.3) there may be no estimation method that leads to a robust estimate of the ATE.

Imbens, 2015

Ensuring Overlap:

1. Matching *without replacement* on the (log-odds ratio of the) propensity score:

$$\hat{l}(x; D, X) = ln\left(\frac{\hat{p}(x; D, X)}{1 - \hat{p}(x; D, X)}\right)$$

   and checking that there is an improvement in the normalised difference. Otherwise, remove some diverging categories.
   Note that $\hat{l}(x; D, X)$ makes it less likely to be matched with an observation which propensity score is close 0 or 1.

2. Trimming: excluding values with extreme values of the propensity score.
   In practice, $\mathbb{A} = \{x \in \mathbb{X} | 0.1 \leq p(X) \leq 0.9\}$ provides a good approximation of the optimal set of preserved values.

Bias adjustment

Covariates are not exactly matched, so difference in average covariates affect estimates.

First step regression:

$$\hat{Y}_{di} = \hat{\alpha}_d + X'\hat{\beta}_d, d = 0, 1 \tag{15}$$

Second step match observations:

$$\hat{X}_{0i} = \begin{cases} X_i & \text{if } D_i = 0 \\ X_{l(i)} & \text{if } D_i = 1 \end{cases}$$

$$\hat{X}_{1i} = \begin{cases} X_{l(i)} & \text{if } D_i = 0 \\ X_i & \text{if } D_i = 1 \end{cases}$$

Bias adjustment

Third step bias adjustment:

$$\hat{Y}_{0i}^{adj} = \begin{cases} Y_i & \text{if } D_i = 0 \\ \hat{Y}_{0i} + \hat{\beta}_0 (X_i - X_{l(i)}) & \text{if } D_i = 1 \end{cases}$$

$$\hat{Y}_{1i}^{adj} = \begin{cases} \hat{Y}_{1i} + \hat{\beta}_1 (X_i - X_{l(i)}) & \text{if } D_i = 0 \\ Y_i & \text{if } D_i = 1 \end{cases}$$

The bias-adjusted matching estimator is then:

$$\widehat{ATE}_M^{adj} = \frac{1}{N} \sum_{i=1}^{M} \left(\hat{Y}_{1i}^{adj} - \hat{Y}_{0i}^{adj} \right)$$

Blocking with regression

Assume a partition of $[0,1]$ into J intervals $[b_{j-1}, b_j]$, for $j = 1, \ldots, J$ with $b_0 = 0$ and $b_J = 1$.

$$B_i(j) = I(b_j - 1 < \hat{p}(X_i) \leq b_j)$$

Then, within each block:

$$\left(\hat{\alpha}_j, \hat{\tau}_j, \hat{\beta}_j\right) = \arg\min_{\alpha,\tau,\beta} \sum_{i=1}^{N} B_i(j) \left(Y_i - \alpha - \tau D_i - \beta' X_i\right)^2 \tag{16}$$

and

$$ATE_{block} = \sum_{j=1}^{J} \frac{N_{cj} + N_{tj}}{N} \hat{\tau}_j \text{ and } ATT_{block} = \sum_{j=1}^{J} \frac{N_{tj}}{N_t} \hat{\tau}_j$$

Blocking with regression

Since the covariates are **balanced within blocks**, the regression does not rely on extrapolation.

The researcher must decide the number of blocks: Imbens provides and algorithm to choose a data-driven number of blocks.

In practice, 5 blocks are usually used.

Imbens, 2015

Conditional variance:

$$\hat{\mathbb{V}}(\hat{\tau}|X,D) = \sum_{i=1}^{N} \left(\frac{D_i}{N_t^2} + \frac{1-D_i}{N_c^2} \right) \lambda_i^2 \hat{\sigma}_{D_i}^2(X_i)$$

with

$$\hat{\sigma_{D_i}^2} = \frac{1}{2}\left(Y_i - Y_{h(i)}\right)$$

and

$$h(i) = \arg \min_{j=1,\ldots,N, j\neq i, D_j=D_i} ||X_i - X_j||$$

Table of Contents

Introduction and Main Assumptions
Identification Results
Parametric Regression Methods
- Regression methods: linear model
- Regression methods: weighted regression
- Doubly-robust estimator

Nonparametric Regression methods
- Introduction to nonparametric regression
- Nonparametric regression for Treatment effect

Matching
A checklist
Extension 1: Counterfactual Analysis
Extension 2: Double/Debiased Machine Learning (DDML)

Selection-on-observables: A Checklist I

This provides you with a series of questions to ask and steps to perform before taking modeling decisions.

1. Classify the variables in your dataset:
   - Which ones are **predetermined** and can serve as controls?
   - Which ones can be thought of as **intermediate outcomes** and serve as explanatory mechanisms or mediators?
   - Is there the possibility to use one for a **placebo test**, e.g. values of the outcome before treatment?
   - Is there really no instrumental variable in my dataset?

Selection-on-observables: A Checklist II

2. Check your data:
   - Always do a data summary.
   - How large is the pool of control?
   - How is the balance of covariates between treatment and controls?

3. Perform a preliminary estimation of the propensity score
   - How does the **distribution propensity score** look like for control and treated?
   - Is there a sufficient region of overlap?
   - Can I estimate the ATE, the ATT, ATNT or a restricted version on a limited support?

Selection-on-observables: Regression and weighted models I

What speaks for a regression model?

1. Linear regression models are the ‘default’. You should always report them. They are viewed as a transparent benchmark and reveal the association in the data.
2. When the data seems sparse in some covariates. For example, minority status may have an effect on the outcome, but it is difficult to find control or treated observations.

When implementing:

1. If manually, demeaning the covariates makes it easier to get at the ATE or ATT quickly.
2. It is not uncommon to add groups of covariates successively. Avoid, if possible, the so-called kitchen-sink approach. It sometimes shows a lack of understanding of the structural equation you are aiming at.
3. The course material presents an example using the R-package ‘marginaleffects’.

Selection-on-observables: Regression and weighted models II

What speaks for a weighted model?

1. IPW is popular because it remains agnostic on the outcome model, which viewed as a 'structural equation'.
2. For the ATT, you should implement when the propensity score is far away from one.
3. Additionally for the ATE, make sure the propensity score is away from zero.

When implementing:

1. Check the **sensitivity to trimming** on the propensity score,
2. Avoid including in the propensity score model covariates that are not confounders (i.e. that don't affect both the treatment status and the outcome),
3. If possible, do doubly-robust estimation.
4. The R material presents manual estimation and estimation with the package 'WeightIt'.

Selection-on-observables: Matching I

What speaks for matching?

1. You have a large enough pool of control variables.
2. Covariates in the original sample are unbalanced (large normalised differences)
3. There are good reasons to expect non-linearities in the outcome variables.

When implementing - follow Imbens (2015):

1. Match without replacement and assess balance.
2. Ensure overlap by restricting the observations to the common support.
3. Re-match with replacement, and use bias adjusted or blocking with regression.
4. Assess and report the sensitivity to your estimation choices (matching criterion, number of neighbours, number of blocks, trimming of the propensity score).

Table of Contents

Introduction and Main Assumptions
Identification Results
Parametric Regression Methods
- Regression methods: linear model
- Regression methods: weighted regression
- Doubly-robust estimator

Nonparametric Regression methods
- Introduction to nonparametric regression
- Nonparametric regression for Treatment effect

Matching
A checklist
Extension 1: Counterfactual Analysis
Extension 2: Double/Debiased Machine Learning (DDML)

Counterfactual Analysis

So far, our interest has been mainly in the treatment effect: $Y_1 - Y_0$, or more precisely, on one feature of its distribution: the average.

We made this choice because the fundamental problem of causal inference makes it difficult to learn about the **joint distribution** of the pair (Y_0, Y_1). Our main tools allow us to make inference on the marginal distribution.

The linearity of the average ensures that the marginal distributions are enough to learn about the average treatment effect.

Counterfactual analysis

The same insight does not work in general: for example, the distribution function or the quantile operator are **not** linear.

Nevertheless, it might still be interesting to go beyond averages and beyond the distribution of individual treatment effects.

A reformulated question can be interesting:

- Instead of: How does the distribution of individual treatment effect look like?
- Ask: How does the treatment affects the distribution of outcomes?

Counterfactual analysis

The two questions share a feature: they ask about a **counterfactual** outcome.

The second question is answerable from the marginal distributions, or the marginal quantiles:

$$\begin{aligned} DE &:= F_{Y_1}(y) - F_{Y_0}(y) = \mathbb{P}(Y_1 \leq y) - \mathbb{P}(Y_0 \leq y) \text{ where } y \in \mathscr{Y} \\ QE &:= Q_{Y_1}(p) - Q_{Y_0}(p), \text{ where } p \in [0,1] \text{ and } Q_{Y_d}(p) := \inf\{y \in \mathscr{Y} : F_{Y_d}(y \leq p)\} \end{aligned}$$

In particular, QE answers both questions if there is **rank invariance**: the treatment does not change my rank in the distribution.

Counterfactual analysis

Note that under the selection on observable hypothesis, both DE and QE are identified.
Note that under unconfoundedness and overlap, the following is also holds:

$$\mathbb{E}(g(Y_d)|X) = \mathbb{E}(g(Y)|X, D = d)$$

for any measurable function g. In particular:

$$F_{Y_d}(y|X) = \mathbb{E}(1\{Y_d \leq y\}|X) = \mathbb{P}(Y_d \leq y|X) = \mathbb{P}(Y \leq y|D = d, X)$$

The marginal distribution is obtained by aggregation over X. Alternatively:

$$F_{Y_1}(y|X) = \mathbb{E}\left(\frac{1\{Y \leq y\} \times D}{\mathbb{P}(D = 1|X)}|X\right) \text{ and } F_{Y_0}(y|X) = \mathbb{E}\left(\frac{1\{Y \leq y\} \times (1 - D)}{\mathbb{P}(D = 0|X)}|X\right)$$

Counterfactual analysis: Chiquiar and Hanson (2005)

Chiquiar and Hanson (2005) take a look at the selection of Mexican migrants to the US, adding to a long debate started by the original contribution of Borjas (1987).

The question of interest: how would the distribution of migrants' wage compare to the distribution of stayers' wage, if they had stayed in Mexico and not migrated to the US?

D: migration status; Y_0: wage income in Mexico, Y_1, wage income in the US.

Counterfactual analysis: Chiquiar and Hanson (2005)

Assuming selection into migration based on observable characteristics (a big assumption!) makes this calculation possible and (fairly) easy.

The observed wage density in Mexico:

$$F_{<0,0>}(y) := \int_{\mathscr{X}_0} F(y|X, D=0) f(x|D=0) dx = \int_{\mathscr{X}_0} F_{Y_0}(y|X) f(x|D=0) dx$$

$F_{Y_0}(y|X)$ captures the wage structure in Mexico (skill prices).
$f(x|D=0)$ captures the observable characteristics of stayers.

Similarly, the observed wage density in the US:

$$F_{<1,1>}(y) = \int_{\mathscr{X}_1} F_{Y_1}(y|X) f(x|D=1) dx \tag{17}$$

Counterfactual analysis: Chiquiar and Hanson (2005)

What we wish to have is:

$$F_{<0,1>}(y) = \int_{\mathscr{X}_1} F_{Y_0}(y|X = x) f(x|D = 1) dx \tag{18}$$

The wage distribution that migrants would have had, had they stayed in Mexico.

Positive selection arises for example if:

$$F_{<0,1>}(y) - F_{<0,0>}(y) \leq 0 \text{ at all } y \in \mathscr{Y}$$

Since selection is only on observable characteristics, this is a matter of adjusting the weights in the distribution. However, nonparametric estimation raises the concern of dimensionality.

Counterfactual analysis: Chiquiar and Hanson (2005)

The trick is to multiply and divide by $f(x|D=0)$.

$$\begin{aligned}
F_{<0,1>}(y) &= \int_{\mathscr{X}_1} F_{Y_0}(y|X)\, f(x|D=1)dx \\
&= \int_{\mathscr{X}_1} F_Y(y|X, D=0)\, \frac{f(x|D=1)}{f(x|D=0)} \times f(x|D=0)dx \\
&= \int_{\mathscr{X}_1} F_Y(y|X, D=0)\, \frac{P(D=1|x)/P(D=1)}{P(D=0|x)/P(D=0)} \times f(x|D=0)dx \\
&= \mathbb{E}_{X|D=0}\left[\mathbb{E}\left(1\{Y \leq y\}\, \theta(X)|X, D=0\right)\right]
\end{aligned}$$

where $\theta(x) := \dfrac{P(D=1|x)/P(D=1)}{P(D=0|x)/P(D=0)}$. The third inequality comes from the Bayes rule: $P(X=x|D=1)P(D=1) = P(D=1|X)P(X=x)$.

Counterfactual analysis: Chiquiar and Hanson (2005)

All that is needed is to re-weight the stayers observation.

$$F_{<0,1>}(y) - F_{<0,0>}(y) = \mathbb{E}_{X|D=0}\left[\mathbb{E}\left(1\left\{Y \leq y\right\}(\theta(X)-1)|X, D=0\right)\right]$$

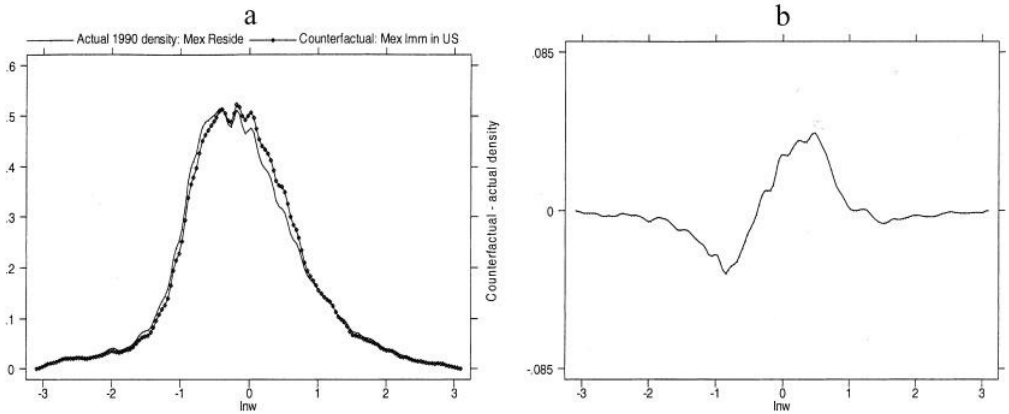
Implementation:

In the same spirit as Inverse Probability weighting:

1. estimate in the first step the propensity score (parametric),
2. then estimate the weights,
3. finally average over the population of interest.

Counterfactual analysis: Chiquiar and Hanson (2005)

Figure: emigrants vs. stayers' wage



Counterfactual analysis: Chernozhukov, Fernandez-Val and Melly (2013)

CFVM offers a semiparametric alternative for calculating the counterfactual distribution:

$$F_{<j,k>}(y) = \int_{\mathscr{X}_1} F_{Y_j}(y|X)\, f_{X_k}(x) dx$$

Instead of estimating the propensity score, we write a flexible model for the distribution of outcome: $F_{Y_j}(y|X) = F_Y(y|X, D = 1)$. For example:

$$F_Y(y|X, D = j) = \Phi\left(R(X)'\beta_j(y)\right)$$

where Φ is the density of the normal distribution, $R(X)$ a polynomial of X (interaction and power term) and $\beta(y)$ varies at each level y.

Counterfactual analysis: Chernozhukov, Fernandez-Val and Melly (2013)

The Distribution regression problem:

$$\hat{F}(Y_i|\boldsymbol{X}_i, D=j) = \Phi\left(R_i'\hat{\pi}_j(y)\right), j = 1, \ldots, J$$

$$\text{where } R_i = r(\boldsymbol{X}_i),\ y \in \mathscr{Y},\ i \in i : D_i = j$$

and

$$\hat{\pi}_j(y) \in \arg\min_{\pi \in \mathbb{R}^{dim(R)}} -\sum_i \left[1\{Y_i^* \leq p\} \log\left(\Phi(R_{ij}'\pi)\right) + 1\{Y_i > p\} \log\left(1 - \Phi(R_{ij}'\pi)\right)\right] \tag{19}$$

Looks complicated, but it is not!

Counterfactual analysis: Chernozhukov, Fernandez-Val and Melly (2013)

Implementation:

1. Slice the support of $\mathscr{Y}$ in L portions: $[y_j, y_{l+1}), l = 1, \ldots, L$ (For example take the deciles).
2. construct $D_{Y,l} = 1\{Y \leq y_l\}$,
3. Perform a generalized linear regression of $D_{Y,l}$ on $R(X)$ for each l, for observations such that $D = j$ and collect the coefficients;
4. Predict: $\hat{F}_Y(y|X, D = j) = \Phi(R(X)'\beta_j(y))$ for observation in group k;
5. Obtain $\hat{F}_{<j,k>}(y)$ by averaging the predicted values over the group k;
6. Inference is done using the bootstrap.

R-package: Counterfactual
`https://journal.r-project.org/archive/2017/RJ-2017-033/RJ-2017-033.pdf`

Table of Contents

Introduction and Main Assumptions
Identification Results
Parametric Regression Methods
- Regression methods: linear model
- Regression methods: weighted regression
- Doubly-robust estimator

Nonparametric Regression methods
- Introduction to nonparametric regression
- Nonparametric regression for Treatment effect

Matching
A checklist
Extension 1: Counterfactual Analysis
Extension 2: Double/Debiased Machine Learning (DDML)

Motivation

Main reference: Chernozhukov et al. (2018)

Regression methods require taking a stand on the functional form of our model e.g., a standard OLS model assumes a linear and additive conditional expectation function (CEF).

We have seen that if one misspecifies the functional form, we will end up with biased estimates of treatment effects even in the absence of unmeasured confounding.

Machine learning (ML) provides a systematic framework for learning the form of the CEF from the data.

Motivation

ML performs much better with high-dimensional data than traditional statistical methods:

1. Many measured covariates (e.g., text data, genetic data);
2. Many non-linear transformations and interactions between these covariates.

ML techniques include: random forests, lasso, ridge, deep neural nets, boosted trees, and various hybrids and ensembles of these methods.

When doing causal inference, we are interested in:

- a low-dimensional parameter θ_0,
- in the presence of high-dimensional nuisance parameters η_0.

Working example

Partially linear model set-up (for illustration):

$$\begin{aligned} Y &= D\theta_0 + g_0(X) + U \\ D &= m_0(X) + V \end{aligned} \tag{20}$$

U and V are the error terms, such that:

$$\mathbb{E}(U|X, D) = 0 \text{ and } \mathbb{E}(V|X) = 0$$

θ_0 is the treatment effect: the parameter of interest.
(g_0, m_0) are the nuisance parameters sometimes referred to as η_0.

The assumption that the effect of D on Y is additive and linear can be relaxed.

Naive ML

LASSO and Ridge for example:

$$\min_{\theta_0,\beta} \quad \left\{\sum_{i=1}^{N}(Y_i - D_i\theta_0 - r(X_i)^T\beta)^2\right\}$$

$$\text{subject to } \sum_{j=1}^{p}|\beta_j| \leq t.\text{(LASSO)}$$
$$\text{subject to } \sum_{j=1}^{p}\beta_j^2 \leq t.\text{(Ridge)}$$

$r(X_i)$ is a (polynomial) vector included power and interaction terms of X_i.

t is a pre-specified free parameter that determines the degree of regularization. It shrinks some parameters to zero.

Choosing the regularization parameter is a fundamental part of LASSO and Ridge: it controls the strength of shrinkage and variable selection, which, in moderation can improve both prediction accuracy and interpretability.

Naive ML

LASSO and Ridge for example:

$$\min_{\theta_0,\beta} \quad \left\{\sum_{i=1}^{N}(Y_i - D_i\theta_0 - r(X_i)^T\beta)^2\right\}$$

$$\text{subject to } \sum_{j=1}^{p}|\beta_j| \leq t.\text{(LASSO)}$$
$$\text{subject to } \sum_{j=1}^{p}\beta_j^2 \leq t.\text{(Ridge)}$$

The shrinkage procedure:

- reduces the variance of the estimator,
- mitigates overfitting, i.e. mistaking noise for signal,
- but introduces a **bias** and prevents $\sqrt{N}$-consistency.

If the regularization becomes **too strong**, important variables may be omitted and coefficients may be shrunk excessively, which can harm predictive capacity and inference.

Naive ML

The example of LASSO/Ridge is symptomatic of other ML procedures that usually suffer from two sources of bias:

1. Bias from regularisation: **introducing some bias** to reduce the variance of the estimator.
2. Bias from overfitting: Fitting the regression too closely to the sample so that we mistake noise for signal. This leads to **poor out-of-sample performance**.

Regularisation bias I

In the following example, the true function is a sinusoid, which we try to approach using a Ridge estimator and a polynomial function.

Regularisation bias II

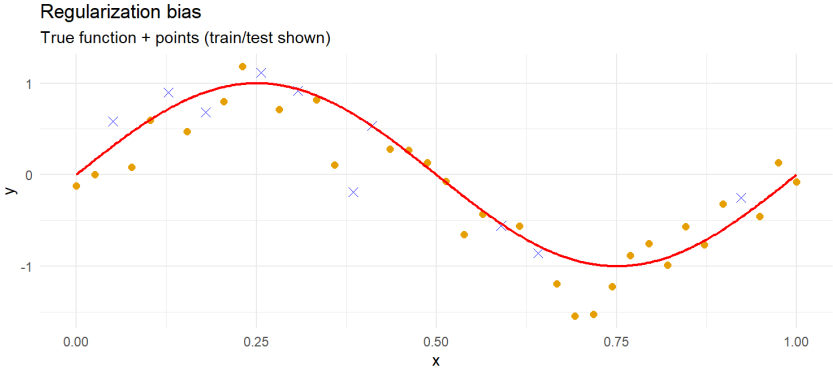


Figure: Illustration of a regularisation bias

Regularisation bias III

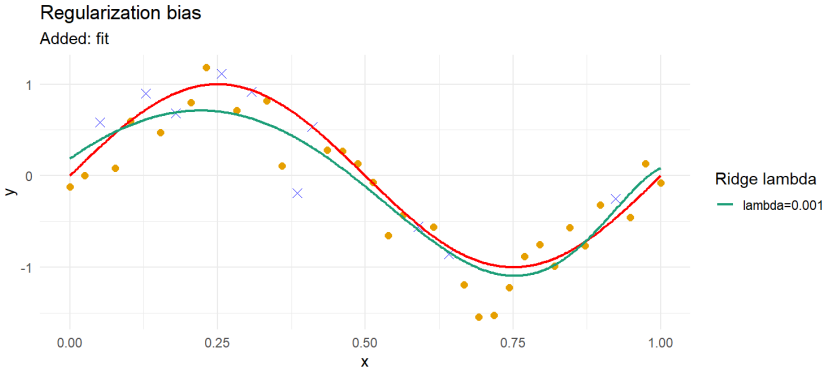


Figure: Illustration of a regularisation bias

Regularisation bias IV

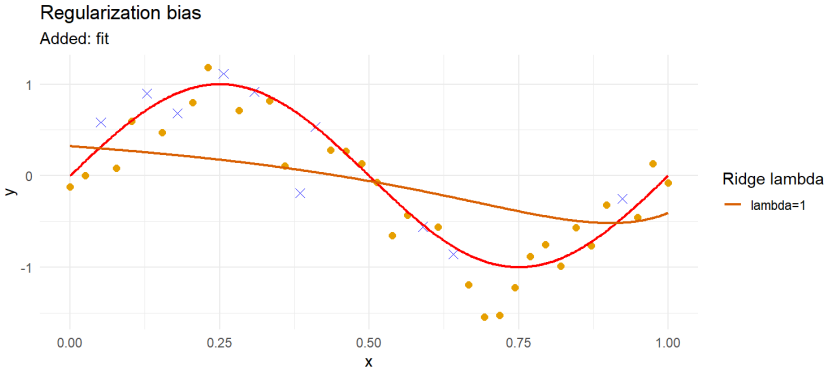


Figure: Illustration of a regularisation bias

Regularisation bias V

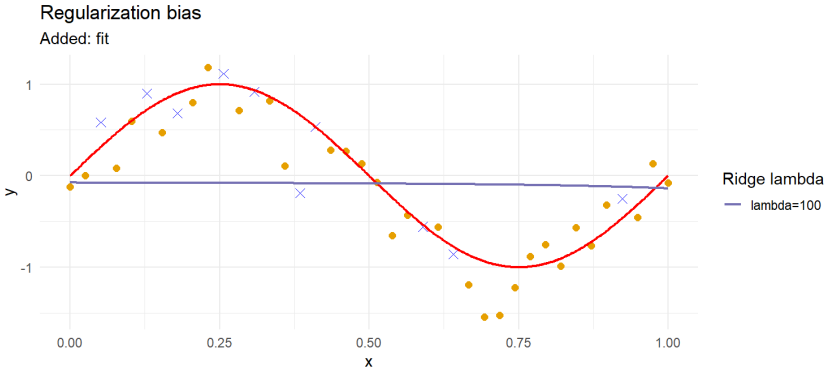


Figure: Illustration of a regularisation bias

Overfitting bias I

Overfitting will occur when we follow too closely the observed points and thus mistake noise for signal.

Overfitting bias II

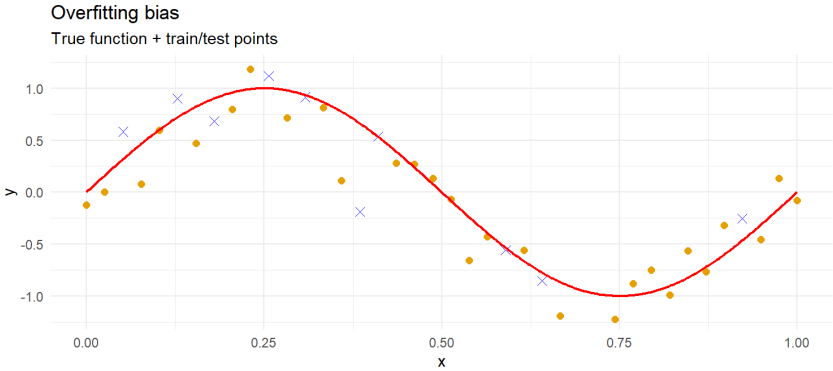


Figure: Illustration of an overfitting bias

Overfitting bias III

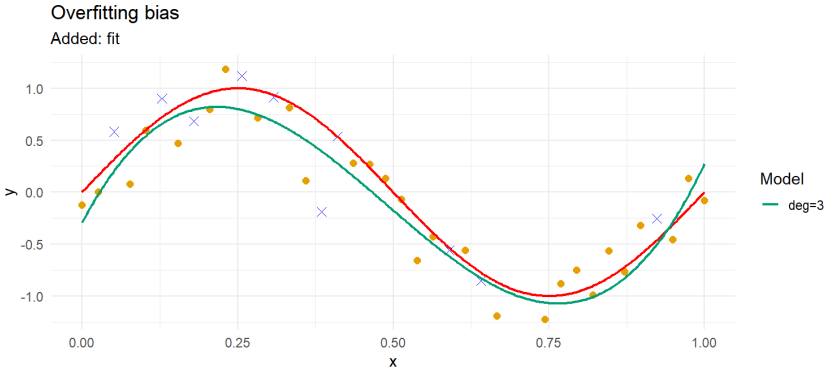


Figure: Illustration of an overfitting bias

Overfitting bias IV

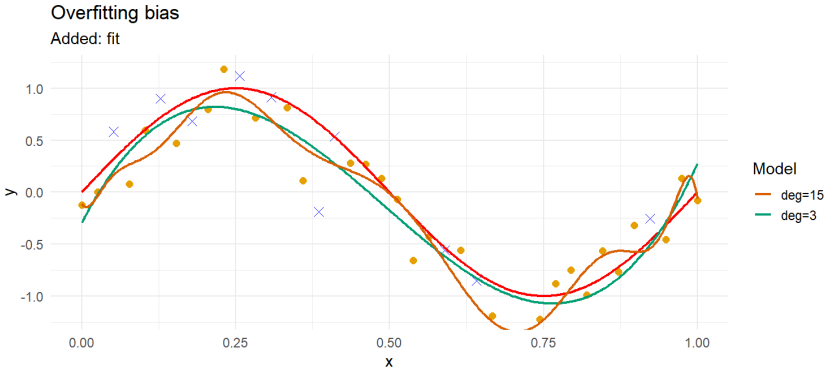


Figure: Illustration of an overfitting bias

Motivation for DDML

We are interested in:

- a low-dimensional parameter θ_0,
- in the presence of high-dimensional nuisance parameters η_0.

The problem with ML:
"ML methods perform well by employing **regularization** to reduce variance and trading off regularization bias with **overfitting** in practice.

However, both regularization bias and overfitting in estimating η_0 cause a heavy bias in estimators of θ_0 that are obtained by naively plugging ML estimators of η_0 into estimating equations for θ_0.

This bias results in the naive estimator failing to be $\sqrt{N}$-consistent, where N is the sample size."

Motivation for DDML

DDML allows us to do causal inference with minimal assumptions about the functional form of our model.

DDML does not help us relax identification assumptions (e.g., no unmeasured confounding, parallel trends, exclusion restriction, etc.)

DDML compared to 'naive ML':

1. Eliminates the bias;
2. Achieves $\sqrt{N}$-consistency;
3. Construct valid confidence intervals

Anatomy of the regularisation bias

Recall that:

$$U_i = Y_i - D_i\theta_0 - g_0(X_i) = D_i\hat{\theta}_0 + \hat{g}_0(X_i) + \hat{U}_i - D_i\theta_0 - g_0(X_i)$$

Isolating and rescaling $(\hat{\theta} - \theta_0)$ gives:

$$\begin{aligned}
\sqrt{N}\left(\hat{\theta} - \theta_0\right) &= \left(\frac{1}{N}\sum_{i=1}^{N} D_i^2\right)^{-1} \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\left(D_i(U_i - \hat{U}_i) + D_i(\hat{g}_0(X_i) - g_0(X_i))\right) \\
&= \left(\frac{1}{N}\sum_{i=1}^{N} D_i^2\right)^{-1} \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\left(D_i(U_i - \hat{U}_i)\right) \quad (:= A) \\
&\quad + \left(\frac{1}{N}\sum_{i=1}^{N} D_i^2\right)^{-1} \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\left(D_i(\hat{g}_0(X_i) - g_0(X_i))\right) \quad (:= B)
\end{aligned}$$

Anatomy of the regularisation bias

A has good properties:

- It has mean zero (unbiasedness);
- It goes to zero as N goes to infinity (consistency);
- It does so at a rate $\sqrt{N}$, so that we can have a CLT.

B does not have good properties:

- It is biased because $\hat{g}_0(X_i)$ is a biased estimator of $g_0(X_i)$;
- It approaches 0 too slowly, at a rate lower than $\sqrt{N}$.

B is the result of the correlation between D and X_i.

ML using orthogonalisation

Instead of fitting one ML model, we fit two models:

1. Treatment model: $D_i = \hat{m}_0(X_i) + \hat{V}_i$
2. Outcome model: $Y_i = \hat{\ell}_0(X_i) + \hat{W}_i$, where $\ell_0(X) = E(Y|X)$

Finally, we regress $\hat{W}$ on $\hat{V}$.

This should remind you of the Frisch-Waugh-Lovell theorem for a linear model:

1. Linear regression of D on X;
2. Linear regression of Y on X;
3. Linear regression of the residuals from 2 on the residuals from 1

ML using orthogonalisation

Instead of fitting one ML model, we fit two models:

1. Treatment model: $D_i = \hat{m}_0(X_i) + \hat{V}_i$
2. Outcome model: $Y_i = \hat{\ell}_0(X_i) + \hat{W}_i$, where $\ell_0(X) = E(Y|X)$

Finally, we regress $\hat{W}$ on $\hat{V}$.

We can call this a 'partialling-out' approach because we have partialled out the associations between X and D and between Y and X.

ML using orthogonalisation

The troublesome term B takes the form:

$$\left(\frac{1}{N}\sum_{i=1}^{N}\hat{V}_i^2\right)^{-1}\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\left((\hat{m}_0(X_i)-m_0(X_i))(\hat{\ell}_0(X_i)-\ell_0(X_i))\right)$$

$(\hat{m}_0(X_i)-m_0(X_i))$: estimation error in the treatment model.
$(\hat{\ell}_0(X_i)-\ell_0(X_i))$: estimation error in the outcome model.

Because this term is based on the product of two estimation errors, it vanishes more quickly!

Bias from overfitting

One can show that the bias also contains a term of the form:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} V_i \left(\hat{\ell}_0(X_i) - \ell_0(X_i) \right)$$

When ℓ_0 is overfit, it will pick up on some of the noise U from the outcome model.

If our noise terms V and U are associated, estimation error in ℓ_0 from overfitting might be associated with V, creating a bias.

Note: $\hat{\ell}_0$ and V might also be associated if we have any unmeasured confounding, but we are assuming that away.

Bias from overfitting

To break the association between V and ℓ_0, we employ sample-splitting:

1. Randomly partition our data into two subsets;
2. Fit your ML models $\hat{\ell}_{0,1}$ and $\hat{m}_{0,1}$ on the first subset;
3. Estimate $\hat{\theta}_{0,1}$ in the second subset using the $\hat{\ell}_{0,1}$ and $\hat{m}_{0,1}$ functions we fit in the first subset.

To preserve efficiency, you can additionally do cross-fitting:

4. Fit an ML models $\hat{\ell}_{0,2}$ and $\hat{m}_{0,2}$ on the second subset;
5. Estimate $\hat{\theta}_{0,2}$ in the first subset using the $\hat{\ell}_{0,2}$ and $\hat{m}_{0,2}$.
6. Average the two estimates of θ_0.

The key idea is to never estimate $\hat{\theta}_0$ using the same observations that used to fit $\hat{\ell}_0$ and $\hat{m}_0$.

Summary

Double/Debiased Machine Learning:

- is useful for selection-on-observables with high-dimensional confounding;
- avoids strong functional form assumptions
- overcomes two sources of bias: regularization and overfitting - two tools for eliminating bias: orthogonalization and cross-fitting.

The formal result, including Neymann orthogonalisation, proof of $\sqrt{N}$-consistency, non-linear models is more involved, and I refer you to the original paper or an easier exposition (inspiration for the slides):
`https://bstewart.scholar.princeton.edu/sites/g/files/toruqf4016/files/bstewart/files/felton.chern_.slides.20190318.pdf`

R-package is self-explanatory: ddml.

Selected Bibliography I

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.

Chernozhukov, V., Fernández-Val, I., and Melly, B. (2013). Inference on counterfactual distributions. *Econometrica*, 81(6):2205–2268.

Chiquiar, D. and Hanson, G. H. (2005). International migration, self-selection, and the distribution of wages: Evidence from mexico and the united states. *Journal of political Economy*, 113(2):239–281.

DiNardo, J., Fortin, N., and Lemieux, T. (1995). Labor market institutions and the distribution of wages, 1973-1992: A semiparametric approach.

Imbens, G. W. (2015). Matching methods in practice: Three examples. *Journal of Human Resources*, 50(2):373–419.

Selected Bibliography II

Imbens, G. W. and Rubin, D. B. (2015). *Causal inference in statistics, social, and biomedical sciences*. Cambridge University Press.